

computer. You can run the `py` command from any folder, because the `C:\Windows` folder is included in the `PATH` environment variable. If you have multiple Python versions installed, running `py` automatically runs the latest version installed on your computer. You can also pass a `-3` or `-2` command line argument to run the latest Python version 3 or version 2 installed, respectively. Or you could enter a more specific version number, such as `-3.6` or `-2.7`, to run that particular Python installation. After the version switch, you can pass all the same command line arguments to `py.exe` as you do to `python.exe`. Run the following from the Windows command line:

```
C:\Users\Al>py -3.6 -c "import sys;print(sys.version)"
3.6.6 (v3.6.6:4cf1f54eb7, Jun 27 2018, 03:37:03) [MSC v.1900 64 bit (AMD64)]

C:\Users\Al>py -2.7
Python 2.7.14 (v2.7.14:84471935ed, Sep 16 2017, 20:25:58) [MSC v.1500 64 bit
(AMD64)] on win32
Type "help", "copyright", "credits" or "license" for more information.
>>>
```

The `py.exe` program is helpful when you have multiple Python versions installed on your Windows machine and need to run a specific version.

Running Commands from a Python Program

Python's `subprocess.run()` function, found in the `subprocess` module, can run shell commands within your Python program and then present the command output as a string. For example, the following code runs the `ls -al` command:

```
>>> import subprocess, locale
❶ >>> procObj = subprocess.run(['ls', '-al'], stdout=subprocess.PIPE)
❷ >>> outputStr = procObj.stdout.decode(locale.getdefaultlocale()[1])
>>> print(outputStr)
total 8
drwxr-xr-x  2 al al 4096 Aug  6 21:37 .
drwxr-xr-x 17 al al 4096 Aug  6 21:37 ..
-rw-r--r--  1 al al   0 Aug  5 15:59 spam.py
```

We pass the `['ls', '-al']` list to `subprocess.run()` ❶. This list contains the command name `ls`, followed by its arguments, as individual strings. Note